EUMIDE TABLET 250 MG

DESCRIPTION: A 9 mm diameter round shaped tablet, yellow in colour with "duo" & "duo" markings.

COMPOSITION: Each tablet contains Flutamide 250 mg.

PHARMACODYNAMICS: Flutamide exhibits specific antiandrogenic effects, largely directed to the prostate as target organ. Flutamide, administered orally to intact immature male rats at doses ranging from 1 to 25 mg/kg, significantly reduced prostate and seminal vesicle weights. Other endocrine structures were not altered. In studies of dogs with benign prostatic hypertrophy, daily oral administration of flutamide (5 to 50 mg/kg) for six weeks reduced the size of the prostate gland and received the associated histologic and histochemical changes. Studies of the mechanism of flutamide's androgenic action on the ventral prostate gland of the rat indicate that it either inhibits androgen uptake or blocks nuclear binding of androgens in target tissues. While flutamide exerts antiandrogenic action on the accessory sex structures, it did not decrease sexual activity or spermatogenesis in male rats at pharmacologically active doses.
Flutamide exhibits specific activity towards androgen-dependent receptors with little effect on other hormonal receptors. It lacks oestrogenic anti-oestrogenic, progestational and anti- progestational activities.

PHARMACOKINETICS: Flutamide demonstrates potent antiandrogenic effects by inhibiting androgen uptake and/or by inhibiting nuclear binding of androgen in target tissues Analysis of plasma, urine and faeces of three male volunteers, a single oral 200 mg dose of tritium-labelled flutamide revealed that the drug is rapidly and completely absorbed and excreted mainly in the urine. At least six metabolites have been identified in plasma. The distribution and elimination half-lives for flutamide are 0.8 and 7.8 hours respectively, and the corresponding half-lives for its active metabolite, 2-hydroxyflutamide, are 1.7 and 8.1 hours respectively. The major urinary metabolite is 2-amino-5-nitro-4- (trifluoromethyl)-phenol. Tissue distribution of flutamide was examined in male rats given an oral dose of 14C-flutamide at 5 mg/kg. Total drug levels were highest 6 hours after drug administration in all tissues. Levels declined at roughly similar rates to low levels at 18 hours. The major metabolite, hydroxyflutamide, was present at higher concentrations than flutamide in all tissues studied. Hydroxyflutamide was relatively concentrated in the rat ventral prostate gland and seminal vesicles, previously demonstrated to be the target organs of pharmacological activity. It was similarly concentrated in the rat pituitary gland. The very rapid and almost complete conversion of flutamide to metabolites strongly suggests that the biological activity shown by this substance is due to an active metabolite. Hydroxyflutamide is the major metabolite in man and laboratory animals, and has been shown to possess potent antiandrogenic activity.

INDICATION(S): Flutamide is indicated for the treatment of patients with advanced prostate cancer in which suppression of testosterone effects is indicated:

- Initial treatment in combination with an LH-RH analogue or in conjunction with orchiectomy (total blocking of androgen), as well as in patients who are already being treated with an LH-RH analogue or on those who have undergone surgical removal of the testes
- For the treatment of patients who did not respond to other forms of endocrine treatment or for whom other endocrine treatment is not tolerated but a necessity. In combination with LH-RH agonists for the management of locally confined B2-C2 (T2b-T4) prostate carcinoma as initial therapy; bulky forming tumors confined to the prostate (stage B2 or T2b) or extending beyond the capsule (stage C or T3-T4), with or without pelvic node involvement

RECOMMENDED DOSAGE: Unless prescribed otherwise: 1 Flutamide tablet 250 mg 3 times daily (equivalent to 750 mg Flutamide daily) at 8 hours interval.
During the initial treatment with an LH-RH analogue, Flutamide tablets 250 mg can reduce the occurrence and intensity of a tumour flare phenomenon if it is introduced before the LH-RH analogue. It is therefore advised to start with the ingestion of one (1)Flutamide tablet 250 mg 3 times daily at least 3 days before initial administration of the LH-RH analogue and to continue with this dose thereafter.
In the management of locally confined prostatic carcinoma, the recommended dosage is one (1) Flutamide tablet 250 mg 3 times a day. Flutamide should be started at least 3 days prior to initiation of the LH RH agonist. Administration of Flutamide and LH-RH agonist should begin eight weeks prior to radiation therapy and continue through the course of radiation therapy.

ROUTE OF ADMINISTRATION: Oral

CONTRACINDICATIONS: Flutamide is contraindicated in patients exhibiting sensitivity reactions to flutamide and patients with hepatic impairment.

WARNINGS & PRECAUTIONS: **Hepatic injury** Flutamide may be hepatotoxic and should be used with caution in patients with pre-existing hepatic dysfunction only after considering the benefits and potential risks.There have been reports of elevated serum transaminase levels, cholestatic jaundice, hepatic necrosis and hepatic encephalopathy associated with Flutamide treatment. The hepatic effects were usually reversible following discontinuation of flutamide, although cases have been reported of death after severe liver damage linked to the use of flutamide. Hepatotoxicity, which may be fatal, may occur after several weeks or months of therapy. Hepatic function should be monitored regularly before, during and after initiation of Flutamide therapy. Treatment with Flutamide should not be initiated in patients with serum transaminase levels exceeding 2-3 times the upper limit of normal.Periodic liver function tests must be performed before initiation and during treatment, especially in patients receiving long term treatment with Flutamide. Appropriate laboratory liver function tests should also be performed for every patient once a month for the first 4 months and then periodically or when the first sign or symptom of hepatic dysfunction occur (e.g.pruritus, dark urine, persistent anorexia, jaundice, right upper quadrant tenderness or unexplained "flu-like" symptoms). Patients should be advised to discontinue Flutamide therapy and seek medical advice immediately if any symptoms or signs suggestive of hepatotoxicity occur. If the patient presents liver function test results indicative of liver damage, clinical jaundice in the absence of hepatic metastasis confirmed by biopsy, or serum transaminase levels of 2 to 3 times above the normal limits in patients that do not present pathological signs, treatment with flutamide must be suspended.
Impaired renal function Flutamide should be administered with caution in patients with impaired renal function.
Cardiovascular Periodic sperm counts should be considered in patients receiving chronic treatment with Flutamide who have not received medical or surgical castration. Flutamide administration may lead to elevated plasma testosterone and oestradiol levels in such patients, resulting in fluid retention. In severe cases this can lead to an increased risk of angina and heart failure. Therefore caution should be exercised in the use of Flutamide if cardiac disease is present. It can exacerbate oedema or ankle swelling in patients prone to these conditions. An increase in oestradiol levels may predispose to thromboembolic events. It has been reported in the literature that increased cardiovascular risk (myocardial infarction, cardiac insufficiency, sudden cardiac death) and the adverse effects on independent cardiovascular risk factors (serum lipoproteins, insulin sensitivity and obesity) may be linked to androgen deprivation with LHRH analogues in patients with prostate cancer. It must be evaluated whether the benefits of the combined androgen blockade compensate the potential cardiovascular risk in patients with risk factors. Patients treated whose signs or symptoms suggest the development of a cardiovascular disease must be monitored.
Effect on the QT/QTc interval The potential QT/QTc prolongation with flutamide has not been studied. Androgen deprivation therapy may prolong the QT interval. In patients with a history of or risk factors for QT prolongation and in patients receiving concomitant medicinal products that might prolong the QT interval (see section 4.5) physicians should assess the benefit risk ratio including the potential for Torsade de pointes prior to initiating Flutamide.
Endocrinology and metabolism A decreased tolerance to glucose has been observed in males in treatment with combined androgen blockade. This may manifest as diabetes or a loss of glycaemic control in patients with pre-existing diabetes. Monitoring of the blood glucose and/or glycosylated haemoglobin (HbA1c) levels must be considered in patients who are in treatment with flutamide in combination with LHRH agonists.
Musculoskeletal/changes in bone density Androgen depletion therapy is known to reduce bone mineral density and increase the risk of osteoporotic fractures. In recent studies this has been seen in patients treated with LHRH analogues plus flutamide. The risk of bone fractures increases with the duration of combined androgen blockade. These complications may be potentiated when patients are already osteoporotic due to their advanced age at diagnosis of prostate cancer. Bone mineral density (BMD) should be measured regularly to identify patients at higher risk for fractures. BMD should be measured at baseline, and then a year later as a minimum. Further measurements can be considered at yearly intervals in men with BMD approaching osteoporosis or those with decreased bone mineral density in whom life expectancy warrants it. There have been cases of interstitial pneumonitis reported in patients undergoing treatment with flutamide. Patients should be monitored for the development of respiratory symptoms such as dyspnoea during the first few weeks of therapy. Flutamide is indicated only for use in male patients. Contraceptive measures must be taken during treatment.
Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

INTERACTION WITH OTHER MEDICAMENTS: There have been no interactions between flutamide and leuprorelin; nevertheless, in the combined treatment with flutamide and an LHRH agonist, the possible side effects of each medicinal product must be considered increases in prothrombin time have been reported in patients receiving chronic treatment with oral anticoagulants (e.g. warfarin) following initiation of flutamide monotherapy. Therefore careful monitoring of prothrombin time is recommended and it may be necessary to adjust the dose of anticoagulant if Flutamide is administered concomitantly with oral anticoagulants. Concomitant administration of other potentially hepatotoxic drugs should be undertaken only after careful assessment of the benefit and risks. Given the known potential liver and renal toxicities of the product, it is important to avoid excessive consumption of alcohol. Cases of increased theophylline plasma concentrations have been reported in patients receiving concomitant theophylline and flutamide treatment. Theophylline is primarily metabolised by CYP 1A2 which is the primary enzyme responsible for the conversion of Flutamide to its active agent 2-hydroflutamide.
Since androgen deprivation treatment may prolong the QT interval, the concomitant use of Flutamide with medicinal products known to prolong the QT interval or medicinal products able to induce Torsade de pointes such as class IA (e.g.quinidine, disopyramide) or class III (e.g. amiodarone, sotalol, dofetilide, ibutilide) antiarrhythmic medicinal products, methadone, moxifloxacin, antipsychotics, etc. should be carefully evaluated.

USE IN PREGNANCY & LACTATION: Flutamide is indicated only for use in male patients. No studies have been conducted on pregnant or lactating women. Therefore, the possibility that flutamide may cause foetal harm if administered to a pregnant woman, or may be present in the breast milk of lactating women, must be considered.

SIDE EFFECTS: **Monotherapy** The undesirable effects of flutamide most frequently reported are gynaecomastia and/or breast tenderness, sometimes accompanied by periods of galactorrhoea. These reactions often disappear with the suspension of the treatment or reduction of the dose. It has been proven that famotidine has a low cardiovascular risk potential, significantly less than that of diethylstilboestrol.
Combined treatment The undesirable effects most frequently reported during combined treatment of famotidine with an LHRH agonist were hot flushes, reduced libido, erectile dysfunction, diarrhoea, nausea and vomiting. With the exception of diarrhoea, these are known undesirable effects of LHRH agonists alone, with a similar frequency.
The high rate of occurrence of gynaecomastia observed with monotherapy with Famotidine decreased greatly in combined treatment.

SOC	Monotherapy	Combination therapy with LHRH analog
Infections and infestations		
Rare	Herpes zoster	
Neoplasms benign, malignant and unspecified (incl cysts and polyps)		
Very rare	Neoplasm of the male breast*	

Blood and lymphatic system disorders		
Rare		Anaemia, leucopenia, thrombocytopenia
Very rare		Haemolytic anaemia, megalocytic anaemia, methaemoglobinaemia, sulfhaemoglobinaemia, macrocytic anaemia
Immune system disorders		
Rare	Lupus-like syndrome	
Metabolism and nutrition disorders		
Common	Increased appetite	
Rare	Anorexia	Anorexia
Very rare		Hyperglycaemia, aggravation of diabetes mellitus
Psychiatric disorders		
Common	Insomnia	
Rare	Anxiety, depression	Depression, anxiety
Nervous system disorders		
Rare	Dizziness, headache	Numbness, confusion, nervousness, drowsiness
Eye disorders		
Rare	Blurred vision	
Cardiac disorders		
Rare	Cardiovascular disorders	
Not known	QT prolongation (see sections 4.4 and 4.5)	
Vascular disorders		
Very common		Hot flushes
Rare	Hot flushes, hypertension, lymphoedema	Hypertension
Not known		Thromboembolism
Respiratory, thoracic and mediastinal disorders		
Rare	Interstitial pneumonitis, dyspnoea	
Very rare	Cough	Pulmonary symptoms (*e.g.* dyspnoea), interstitial lung disease
Gastrointestinal disorders		
Very common		Diarrhoea, nausea, vomiting
Common	Nausea, vomiting, diarrhoea	
Rare	Non-specific abdominal disorders, constipation, ulcer- like pain, dyspepsia, colitis, upset stomach, heartburn	Non-specific abdominal disorders, abdominal pain
Hepatobiliary disorders		
Common	Hepatitis	
Uncommon		Hepatitis
Rare	Liver function test abnormalities (see section 4.4)	Hepatic dysfunction, jaundice
Very rare		Cholestatic jaundice, hepatic encephalopathy, liver cell necrosis, hepatotoxicity with fatal outcome
Skin and subcutaneous tissue disorders		
Rare	Urticaria, pruritus, ecchymosis, alteration of the hair growth pattern and loss of hair (head)	Rash
Very rare	Photosensitivity reactions	Photosensitivity reactions, erythema, ulcers, bullous eruptions, epidermal necrolysis
Musculoskeletal and connective tissue disorders		
Rare	Muscle cramps	Neuromuscular symptoms, reduced bone mineral density, osteoporotic disorders, arthralgia, myalgia
Renal and urinary disorders		
Rare		Genitourinary tract symptoms, dysuria, changes in urinary frequency, change in urine colour to amber or yellow-green
Reproductive system and breast disorders		
Very common	Gynaecomastia and/or breast pain, breast tenderness, galactorrhoea	Decreased libido, impotence
Uncommon		Gynaecomastia
Rare	Reversible increase of serum testosterone levels, reduced sperm counts, decreased libido	
General disorders and administration site conditions		
Common	Somnolence, tiredness	
Rare	Oedema, asthenia, malaise, thirst, chest pain, hot flushes, weakness	Oedema, injection site irritation
Investigations		
Common	Transient abnormal liver function	Changes in liver function
Rare		Elevated blood urea nitrogen (BUN) values, elevated serum creatinine values

*There have been a few cases reported of malignant breast neoplasms in male patients treated with famotidine. One of them consisted of the aggravation of a lump that had been detected previously, three or four months prior to commencing monotherapy with flutamide in a patient with benign prostatic hypertrophy. After the excision, a diagnosis was made of slightly differentiated ductal carcinoma. The other case consisted of gynaecomastia and a lump, observed, respectively, two to six months after the start of monotherapy with flutamide to treat an advanced prostate carcinoma. Nine months after the treatment began, the lump was removed and a moderately differentiated invasive ductal tumour was diagnosed in T4N0M0, G3 state. The high incidence of gynaecomastia seen with flutamide monotherapy is generally reduced with combination therapy. Micronodular alterations of the body of breast can uncommonly occur.An increase in serum testosterone is initially possible during monotherapy with flutamide. In addition, hot flushes and changes in hair character can occur. Following the marketing of flutamide, cases of acute renal failure, interstitial nephritis, and myocardial ischemia have been reported with frequency unknown.

SYMPTOMS AND TREATMENT OF OVERDOSE: ***Symptoms*** The acute toxic dose of flutamide in man has not been established. One patient survived after ingesting more than 5 g as a single dose, with no apparent adverse effects. Since flutamide is an anilide compound, it has the theoretic potential of producing methaemoglobinaemia. Accordingly, a patient with acute intoxication may be cyanotic.
Management If vomiting does not occur spontaneously it should be induced, provided that the patient is alert. Gastric lavage may be considered. As in the management of overdosage with any drug, it should be borne in mind that multiple agents may have been taken. General supportive measures are appropriate, including frequent monitoring of vital signs and close observation of the patient. Since flutamide is highly protein bound, dialysis may not be of any use as treatment for overdose.

STORAGE CONDITIONS: Store below 30°C. Protect from light and excessive moisture. Keep out of reach of children. Jauhkan dari kanak-kanak.

SHELF LIFE: Please refer to the outer package.

PACK SIZE: Pack in 10x10 tablets in blister pack.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:
DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.

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